



ScienceDirect

① The George Washington University does not subscribe to this content on ScienceDirect.

Measurement

Date: January 2022

[Show more](#) ✓

Research article [Get rights and content](#) ↗

A CNN-SVM study based on selected deep features for grapevine leaves classification

Murat Koklu ^a , M. Fahri Unlarsen ^b , Ilker Ali Ozkan ^a , M. Fatih Aslan ^c , Kadir Sabanci ^c [Show more](#) ✓

Cite Add to Mendeley Share | 10.1016/j.measurement.2021.110425 ↗

[Access options](#)

Article preview

Recommended articles

Cited by (159)

Metrics

Highlights

- Classification of five classes of grapevine leaves by MobileNetv2 [CNN](#) Model.
- Classification of features using SVMs with different kernel functions.
- Implementing a feature selection algorithm for high classification percentage.
- Classification with highest accuracy using CNN-SVM Cubic model.

Article preview

Abstract

Introduction

Section snippets

References (62)

Abstract

The main product of grapevines is grapes that are consumed fresh or processed. In addition, grapevine leaves are harvested once a year as a by-product. The species of grapevine leaves are important in terms of price and taste. In this study, deep learning-based classification is conducted by using images of grapevine leaves. For this purpose, images of 500 vine leaves belonging to 5 species were taken with a special self-illuminating system. Later, this number was increased to 2500 with data augmentation methods. The classification was conducted with a state-of-art [CNN](#) model fine-tuned MobileNetv2. As the second approach, features were extracted from pre-trained MobileNetv2's Logits layer and classification was made using various SVM kernels. As the third approach, 1000 features extracted from MobileNetv2's Logits layer were selected

by the Chi-Squares method and reduced to 250. Then, classification was made with various SVM kernels using the selected features. The most successful method was obtained by extracting features from the Logits layer and reducing the feature with the Chi-Squares method. The most successful SVM kernel was Cubic. The classification success of the system has been determined as 97.60%. It was observed that feature selection increased the classification success although the number of features used in classification decreased.

Keywords

Deep learning; Transfer learning; SVM; Grapevine leaves; Leaf identification

[< Previous article in this issue](#)

[Next article in this issue >](#)

Introduction

Grapevine leaves are a type of leaf that is often used in the traditional food culture of Turkish cuisine by processing it brine, canned and frozen [1]. In some grape species, grapevine leaves are more expensive than fruit. Vine varieties have leaves that show very different characteristics in terms of criteria such as shape, thickness, featheriness, sliciness. For this reason, the leaves of each variety are not used for cooking. Thick, feathered, and over-sliced leaves are not preferred by consumers. It is preferred that the vine variety to be used for cooking is thin, featherless, thin-veined, as sliced as possible and leaves a sour taste on the palate. [2], [3]. Therefore, the separation of edible species from other vine species and the determination of vine species from leaf and fruit images is an important requirement in this area. However, for non-specialists, it is quite difficult to determine the type from grapevine leaves.

Fig. 1 shows five types of grapevine leaves commonly consumed in the Central Anatolia region of Turkey. The shape and structure of grapevine leaves also differ between edible vine species of different quality [4], [5]. As can be seen from the figure, although differences in shape and structure appear in different edible species, genetic variations within the species, as well as environmental factors such as light and water, can affect the shape of grapevine leaves [6]. In addition, the leaf shapes of grapevines may vary in the growth process [7]. Some species have high similarities between each other, and it can take a long time to distinguish them. For these reasons, it is very difficult to classify all the edible vine types for an expert and is not practical. Because of all these factors, different methods and techniques are needed to distinguish between edible Grapevine leaves varieties that have economic value.

Currently, the increase in artificial intelligence applications in the field of agriculture brings forth solutions to problems in this field and creates an alternative to the methods used until today [8]. In recent years, plant diagnostic systems have been successfully used to solve problems such as yield, disease, species estimation [9], [10], [11]. There is still a need to work on the

identification of visual data obtained from the plants in the field of agriculture [12]. Fruit, flowers, and leaves are classified by using visual data. Flowers and fruits cannot be used in plant identification problems for long-term, as they appear over a limited period of time. However, the use of leaves are more suitable for plant identification due to the fact that they contain plant-specific characteristics, are numerous and are available most of the year [13], [14], [15].

In recent years, many researchers have developed leaf image analysis and machine learning techniques for plant classification [16], [17], [18], [19]. Formal characteristics of leaves are used to describe and classify plants. [20]. In these studies, morphological features of leaf images are extracted by image processing and the feature extraction process is performed [17], [18], [21], [22], [23]. When the literature is examined, it is observed that the extraction of a small number of features limits identification performance. In order to achieve high accuracy, studies are needed on methods that provide deeper information on the leaves of plants.

Deep learning approaches have become a newer technique in the field of computer vision and can extract more detailed information compared to machine learning techniques [24], [25], [26], [27], [28], [29]. Deep learning approaches have come to the fore in recent years with their ability to automatically extract attributes. Deep learning is usually evaluated as a black-box approach. In this approach, images are given as input, and deep learning automatically determines features. Then, classification is performed with these features [30]. In recent years, various methods of plant leaf classification based on deep learning have been proposed [31], [32], [33], [34], [35], [36], [37], [38], [39].

Edible grapevine leaves contain many formal characteristics, and it is required to classify grapevine leaves with the most economic value through these characteristics. The separation of consumable grapevine leaves according to its quality also has an important place in marketing. The aim of this study is to develop a diagnostic system for grapevine leaves, which require expert knowledge and are quite difficult to distinguish visually. In this field, there are generally studies on determining the diseases of grapevine leaves, but on the other hand, manual measurement techniques have been used in the classification of grapevine leaves until now. In this study, three systems that automatically classify the grapevine leaves according to its type are proposed to address the mentioned problems. There are studies on the classification of different plants based on leaves. The studies are few in number and it is difficult to classify the leaf species because their characteristics are highly similar. Also, when the studies in the literature using leaf images are examined, it is seen that they are generally on disease and different plant species classification. Considering the studies on grape leaves, although there are studies on the classification of diseases, there is still no study to distinguish different grape leaf species belonging to the same region.

In this respect, this study differs from other studies in the literature. In the first system, species classification was made with a current CNN model. In the second

system, the features were extracted from the CNN model and classified with SVM algorithms. In the third system, the determined portion of the features extracted from the CNN model with the feature selection is classified by SVM algorithms. The performance of the three models is compared.

The contributions of our study can be briefly summarized as follows:

- Classification of five classes of grapevine leaves by fine-tuned MobileNetv2 CNN Model
- Classification of features extracted from pre-trained MobileNetv2 models using SVMs with different kernel functions, i.e. CNN-SVM structure,
- Implementing a feature selection algorithm for high classification percentages.

This work successfully combines several concepts, approaches, techniques, and components, such as Image Acquisition, Image augmentation, Image classification, Feature Extraction, Deep Learning, Transfer learning, SVM, Grapevine leaves, Leaf identification. This is a typical combination novelty, which can be highlighted to show the contributions and/or advantages of the proposed method.

In literature, there are various studies about disease estimation and species classification of plants by using leaf images and employing different artificial intelligence methods so far. These classification studies have generally been used in the identification of leaf diseases. A comparative summary of the studies found in the literature based on leaf images is given in Table 1.

Barré et al. proposed a deep learning model, called LeafNet, where distinguishing features can be learned from leaf images. They stated that LeafNet revealed a better performance than traditional methods in different data [40]. Saini et al. has proposed a deep learning technique with the help of the deep CNN instead of traditional classification methods in the classification of plant leaves. They have shown that feature extraction is performed both automatically and in a short time with this method, and they have also achieved high accuracy [34]. Lee et al. has concluded that leaf shape is not enough by itself to recognize plant species among closely related species. It has been implied that extracted features using CNN show better results compared to handcrafted features [33]. Tan et al suggested the usage of a pre-trained CNN model for the feature extraction of 43 different plants' leaves. They have classified the extracted features from the suggested model using SVM, ANN, k-NN and NB machine learning algorithms. Compared to classical morphometric measurements which gave 66.55% accuracy, this method provided 94.88% accuracy [24]. There are also different papers about illness detection and recognition on grapevine leaves using deep learning methods [41], [42], [43].

There is no deep learning study to distinguish grapevine leaves species. Paravithana and Kalansuriya employed CNN to predict the suitability of tea

buds for the plucking in their study. They photographed the suitable and unsuitable tea buds. It is reported that the CNN model has a classification accuracy of 70.15%. For the same dataset, the SVM and Inception V3 methods have 65.86% and 68.70% respectively [44]. Naik and Shah took a total of 50,000 images of the leaves of 50 plant species growing in Southern Gujarat. Pre-trained models of CNN (Inception V4, Xception, ResNet, InceptionResNetV2, DenseNet, MobileNet) are employed for feature extractor. These features are classified by using the logistic regression machine learning method. In experiments, Xception has the best performance with an accuracy of 93.4% [45]. Hasan et al. proposed in their study a grape leaf diseases identification method based on CNN. The python Keras library is used in this article. The performance of the method is presented as 91.37% in terms of accuracy [46]. Ji et al. proposed a CNN architecture to distinguish the grape leaf diseases (black rot, esca and isariopsis leaf spot), and healthy leaf. Multiple CNNs combinations are presented in this article. The PlantVillage dataset is used for comparison of well known CNN models. The proposed structure archives 98.57% accuracy in tests [42]. Liu et al. proposed a CNN based leaf diseases diagnosis method. Totally 107,366 leaf image is created by augmenting 4023 newly taken images and 3646 images collected from public datasets. It is reported that the proposed DICNN has 97.22% accuracy which is bigger than GoogLeNet and ResNet-34 with the ratio of 2.97% and 2.55% respectively [47]. Ghoury et al. proposed a disease detection method for grape and grape leaf. SSD_MobileNet v1 and Faster R-CNN Inception v2 models have been employed for this purpose. The classification accuracy of the Faster R-CNN Inception v2 model is between 78% and 99%. It is reported that the image resolution, background change and noise increase the misclassifications [48].

Unlike the above studies, our study creates a grapevine leaf dataset and this leaf type has not been studied before. In terms of methodology, the proposed CNN-SVM structure and feature selection steps differ from previous studies.

Access options

 [Find It @ Himmelfarb](#)

 [Find It @ GW Libraries](#)

[Access through another organization](#)

[Need help with access? ↗](#)

Section snippets

Methodology

In this section, the methods used in the classification of grapevine leaves were presented in detail. The stages of the proposed methods and the order of these stages are presented in Fig. 2. First, an image acquisition system was prepared to take images of grapevine leaves and a dataset was created. The image acquisition system and its key properties are described in the next subsection. Various image augmentation methods such as rotation, scaling, translation, etc. have been applied to ...

Results and discussion

In this section, the success of the three methods created using the state-of-the-art MobileNetv2 proposed for grapevine leaves classification is presented. MobileNetv2 is a pre-trained CNN model that uses images with 224x224x3 resolution and creates 1000 classes. A computer with an Intel core i7-7700HG processor, NVIDIA GeForce GTX 1050 4GB display card and 16GB of RAM is employed in the deep learning experiments in this study. In this study, three different methods based on state-of-art CNN ...

Conclusion

Grapevine leaves are classified using three different methods using MobileNetv2, one of the popular CNN models. Background color has been chosen as black for the leaves to be easily distinguished from the background in the images. A total of 500 images are obtained by taking 100 images for each type. The number of the grapevine image has been increased up to 2500 using different data augmentation techniques. Thus, sufficient data amount is reached for deep network training. Three different ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

References (62)

A. Sicard *et al.*

[Repeated Evolutionary Changes of Leaf Morphology Caused by Mutations to a Homeobox Gene](#)

Curr. Biol. (2014)

D.I. Patrício *et al.*

[Computer vision and artificial intelligence in precision agriculture for grain crops: a systematic review](#)

Comput. Electron Agric. (2018)

Z. Li *et al.*

[A review of computer vision technologies for plant phenotyping](#)

Comput. Electron. Agric. (2020)

R. Manavalan

[Automatic identification of diseases in grains crops through computational approaches: A review](#)

Comput. Electron. Agric. (2020)

A. Aakif *et al.*

[Automatic classification of plants based on their leaves](#)

Biosyst. Eng. (2015)

M. Koklu *et al.*

[Multiclass classification of dry beans using computer vision and machine learning techniques](#)

Comput. Electron. Agric. (2020)

E. Kaya *et al.*

[Towards a real-time sorting system: Identification of vitreous durum wheat kernels using ANN based on their morphological, colour, wavelet and gaborlet features](#)

Comput. Electron. Agric. (2019)

J.-X. Du *et al.*

[Leaf shape based plant species recognition](#)

Appl. Math. Comput. (2007)

J. Chaki *et al.*

[Fragmented plant leaf recognition: Bag-of-features, fuzzy-color and edge-texture histogram descriptors with multi-layer perceptron](#)

Optik. (2019)

M.F. Aslan *et al.*

[CNN-based transfer learning–BiLSTM network: A novel approach for COVID-19 infection detection](#)

Appl. Soft Comput. (2021)



View more references

[View full text](#)

Recommended articles

Based on reading popularity

Research article

[Ensemble of heterogeneous classifiers for diagnosis and prediction of coronary artery disease with reduced feature subset](#)

Durgadevi Velusamy, Karthikeyan Ramasamy

Computer Methods and Programs in Biomedicine • Volume 198 • 2021 • Article 105770

Research article

[An experimental study on the effects of adding biomass ashes on ash sintering behavior of Zhundong coal](#)

Hao Zhou, Weichen Ma

Applied Thermal Engineering • Volume 126 • 2017 • Pages 689-701

Research article

Image processing based characterisation of coal cleat networks

J. Busse, ..., A. Scheuermann

International Journal of Coal Geology • Volume 169 • 2017 • Pages 1-21

Research article

Fabrication of chitosan/graphene oxide-gadolinium nanorods as a novel nanocomposite for arsenic removal from aqueous solutions

Jong-Soo Choi, ..., Janardhan Reddy Koduru

Journal of Molecular Liquids • Volume 320, Part A • 2020 • Article 114410

Research article

A progressive failure analysis of all-C/SiC composite multi-bolt joints

Libin Zhao, ..., Jianyu Zhang

Composite Structures • Volume 202 • 2018 • Pages 1059-1068

Research article

Vibration impact of rock excavation on nearby sensitive buildings: An assessment framework

Shiguang Wang, Songye Zhu

Soil Dynamics and Earthquake Engineering • Volume 163 • 2022 • Article 107508

Cited by (159)

A systematic review of deep learning techniques for plant diseases ↗

Pacal I., ..., Martinek R.

Artificial Intelligence Review • Volume 57 • 2024 • Article 304

Advancements in deep learning for accurate classification of grape leaves and diagnosis of grape diseases ↗

Kunduracioğlu I., Pacal I.

Journal of Plant Diseases and Protection • Volume 131 • 2024

Underwater Target Detection Based on Improved YOLOv7 ↗

Liu K., ..., Wang N.

Journal of Marine Science and Engineering • Volume 11 • 2023 • Article 677

Identification Method of Corn Leaf Disease Based on Improved Mobilenetv3 Model ↗

Bi C., ..., Yu H.

Agronomy • Volume 13 • 2023 • Article 300

A New Dawn for Tomato-spotted wilt virus Detection and Intensity Classification: A CNN and LSTM Ensemble Model ↗

Sharma R., ..., Vats S.

2023 4th International Conference for Emerging Technology Incet 2023 • 2023

Deep Learning Based Detector YOLOv5 for Identifying Insect Pests ↗

Ahmad I., ..., Zhang Y.



View all citing articles on Scopus ↗

Metrics

Citations	
Citation Indexes	159
Captures	
Mendeley Readers	154
 View details ↗	

© 2021 Elsevier Ltd. All rights reserved.



All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

